

Datasheet

TarangMini™ ST22LR01



REVISION HISTORY

Version	Date	Section	Changes
0.1	20-04-2021	All	Draft - Initial Release
0.4	18-05-2021	5	Increased the tolerance value
1.0	09-09-2021	All	Updated
1.1	12-05-2022	10	Updated soldering recommendation
1.2	24-05-2023	All	Updated

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1.PRODUCT DESCRIPTION

Melange Systems TarangMini™ ST22LR01 Sub-GHz LoRa® wireless module is designed to fulfil the last mile connectivity requirements of IoT applications. TarangMini™ST22LR01 series module is built around STMicroelectronics' STM32 platform and powered by an ARM® Cortex® M4 processor. With Melange Systems Tarang™ software eco system in place, users get the flexibility to choose between multiple software stacks appropriate for their application. These cost effective, low power modules are built in a small form-factor and has very simple interfacing options to provide wireless capabilities to any existing embedded products. TarangMini™ST22LR01 module has provision to mount different antenna types and provides SMD or through-hole mounting options to a base-board

HIGHLIGHTS

- Completely integrated module with no external component requirement
- Ready network stack binaries for faster time-to-market; no expertise required
- Simple UART Commands to configure and control the module
- ARM® Cortex® M4 processor
- Integrated LoRa® and GFSK Radio
- Programmable output power up to +22 dBm
- Supports Melange Systems' TarangConnect - LW-S201 (LoRaWAN® Binary Class A & Class C)
- Supports Melange Systems' TarangConnect - TN-S101 (module to module)*
- Integrated 64 Bit EUI
- Provision to mount on-board helical antenna, ext antenna and pad to connect to base board
- Provision to mount as an SMD or through hole daughter card to the base board
- Optimised for IN865-867 License free band
- WPC - ETA Regulatory Certified for legally operating in India
- Firmware customization and additional certification support for bulk purchases

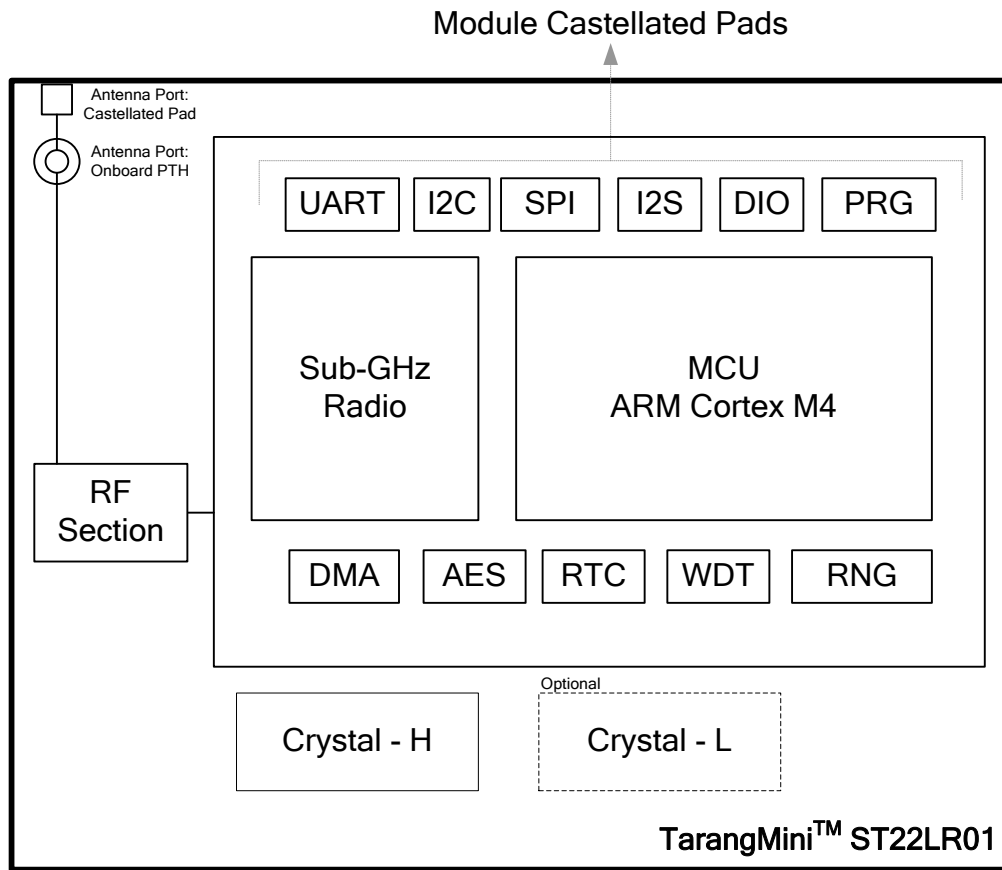
TYPICAL APPLICATIONS

- Street Lighting Automation
- Smart Energy Metering
- Smart Water Metering
- Vehicle / Asset Tracking
- Smart Parking Solution
- Smart City Applications
- Agricultural Application
- Industrial Automation

ORDERING INFORMATION

Ordering Part Number	Description
TarangMini ST22LR01-STD-TR	LoRa Module - Tape and Reel: SPQ 250 Numbers
TarangMini ST22LR01-STD-SP	LoRa Module Samples: Cut-tape/ ESD pouch
AH866T11100TV090-T	(Opt)On-board vertical helical antenna - Tray: SPQ 100 Numbers
AH866T11100TV090-S	(Opt) On-board vertical helical antenna Samples: Loose packet
TarangMini USB-EVK	(Opt) USB interface card with cable and AC power adaptor

2.FUNCTIONAL BLOCK DIAGRAM



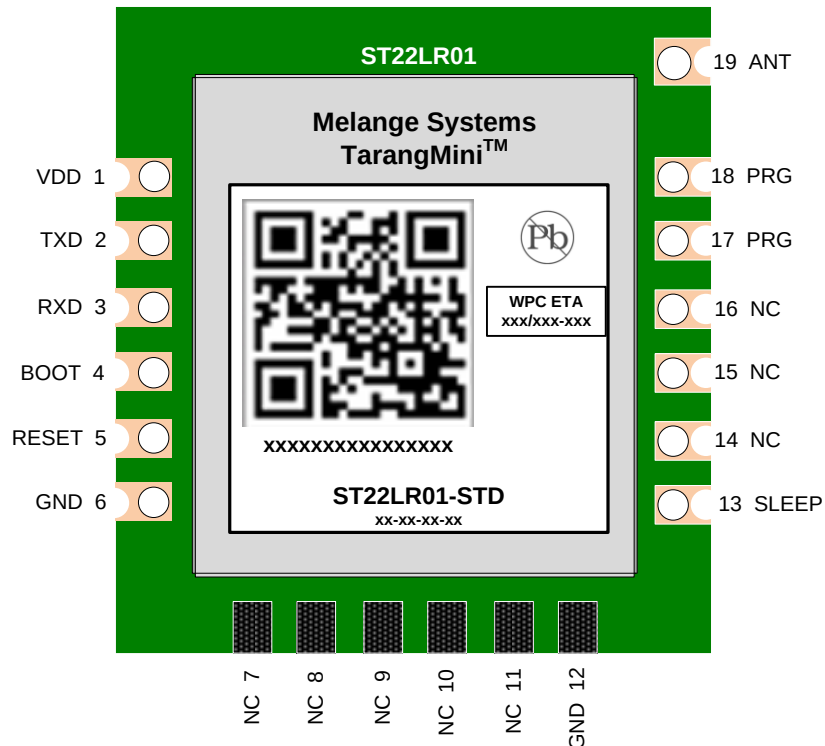
Note 1: Default firmware binaries are enabled with single UART peripheral and a sleep pin to interface with any host controller

Note 2: Default firmware binaries may use other internal peripherals as per the network stack requirements

Note 3: Refer respective firmware documentations for more details

Note 4: Bulk purchases are supported with firmware customisations utilizing other peripherals also. Contact Melange Systems (iot.support@melangesystems.com) for such requirements

3. MODULE PIN DESCRIPTION



Pin	Name	Type	Function
1	VDD	IN	Power Supply
2	TXD	OUT	UART Data Out
3	RXD	IN	UART Data In
4	BOOT	IN	H-Firmware Update; L-Normal Operation
5	RESET	IN	System Reset (Active Low)
6	GND		Ground
7	NC		Reserved (Bottom Side Pad Only)
8	NC		Reserved (Bottom Side Pad Only)
9	NC		Reserved (Bottom Side Pad Only)
10	NC		Reserved (Bottom Side Pad Only)
11	NC		Reserved (Bottom Side Pad Only)
12	GND		Ground
13	SLEEP	IN	Sleep Control Line
14	NC		Reserved
15	NC		Reserved
16	NC		Reserved
17	PRG		Programming Pin (Reserved)
18	PRG		Programming Pin (Reserved)
19	ANT		RF Port (for optional external antenna)

4. SPECIFICATIONS

RADIO

Specification	Description
Frequency Band	863 MHz to 870 MHz
Modulation Scheme	LoRa®, (G)FSK
Maximum Output Power	+21.5 dBm
Receiver Sensitivity	-134 dBm @ 1% PER

Operating Conditions: VDD 3.3V, FREQ: 865.4025MHz, SF12, BW: 125KHz, RFOP set @ 22 dBm, Temperature: +25°C

GENERAL

Specification	Description
Processor	Arm® Cortex M4, 48MHz
Host Interface	UART
Default Firmware	TarangConnect - LW-S201 (LoRaWAN® Class A & C)
Default Channels	Tx: 865.0625 MHz, 865.4025 MHz, 865.9850 MHz, Rx:866.55 MHz
Default Bandwidth	125 KHz
Bitrate	DR0 - DR5 (0.3 - 5 kbps)
Output Power Configuration	0 dBm to 22 dBm (Step Size: 1)
Operating Temperature	-40 to +85 °C
Storage Temperature	-40 to +85 °C

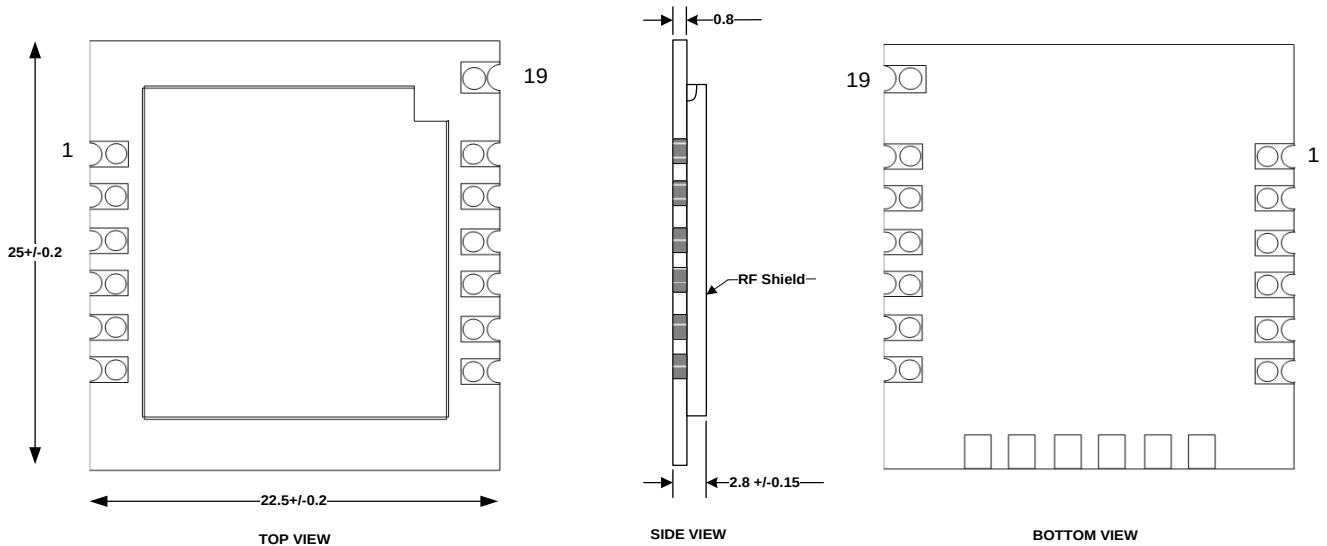
5. ELECTRICAL CHARACTERISTICS

Specification	Minimum	Typical	Maximum	Unit
Operating Voltage (VDD)	1.8	3.3	3.6	V
Voltage on any other pins	VSS - 0.3		3.6	V
Logic Input Voltage (High)	0.7 x VDD			V
Logic Input Voltage (Low)			0.3 x VDD	V
Brownout Reset Voltage	1.72	1.76	1.8	V
GPIO Sink Current			±20	mA
Transmit Mode Current	113	-	130	mA
Receive Mode Current	7	-	10	mA
Idle Mode Current	3	-	7	mA
Sleep Mode Current	2	-	4	uA

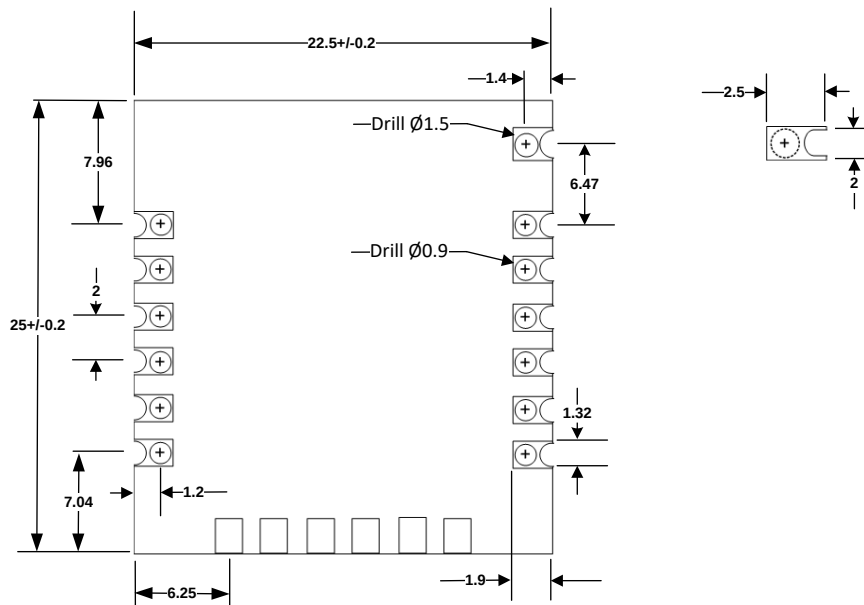
Operating Conditions: VDD 3.3V, FREQ: 865.4025MHz, SF12, BW: 125KHz, RFOP @ 22 dBm, Temperature: +25°C, FW: TarangConnect - LW-S201

6.MECHANICAL SPECIFICATIONS

MODULE DIMENSIONS



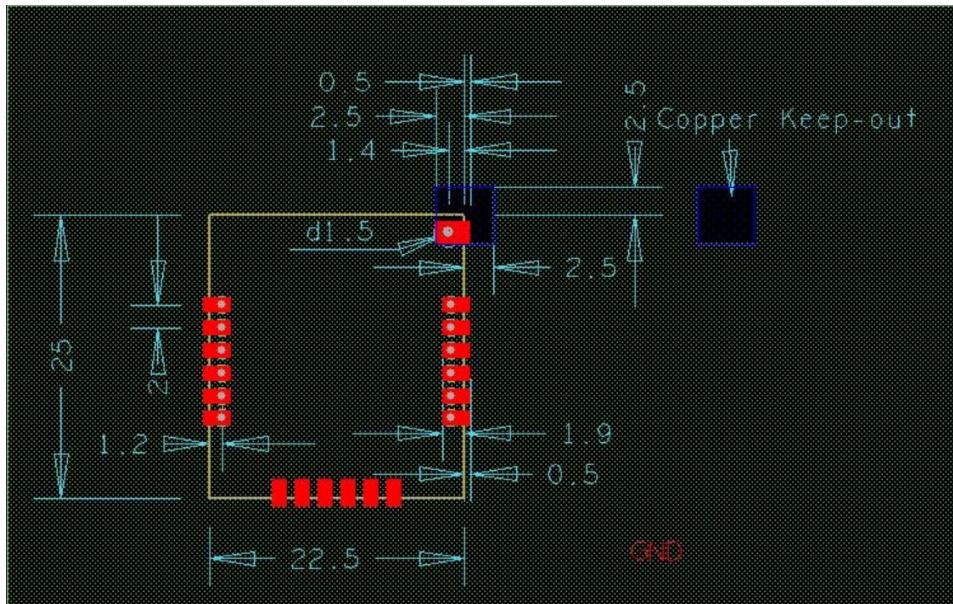
MODULE FOOTPRINT



ALL DIMENSIONS ARE IN MILLIMETRE

7.PCB DESIGN RECOMMENDATIONS

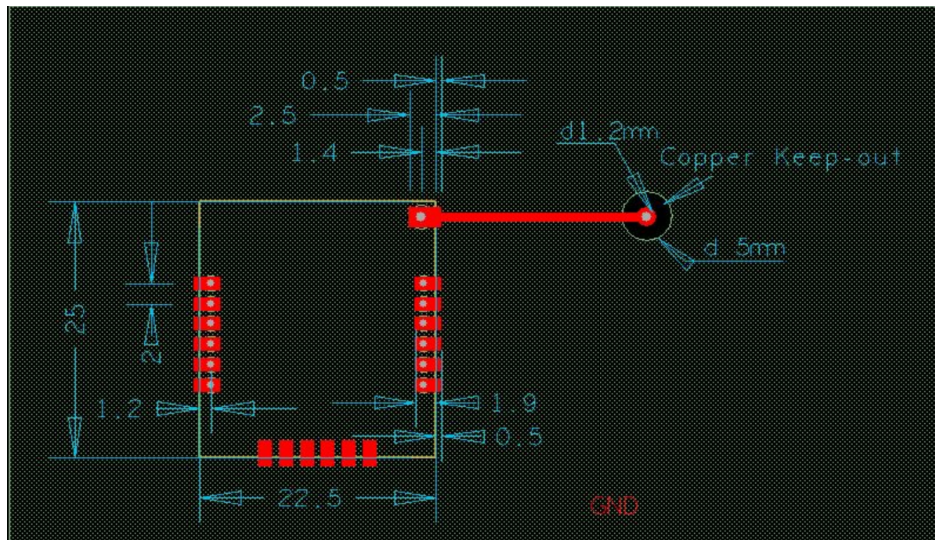
HOST PCB LAND PATTERN &KEEP-OUT FOR ON-BOARD ANTENNA



Note:

- We recommend to provide a PCB cut-out at the copper keep-out area of the host PCB if an on-board spring antenna is used (for through-hole soldering).
- Refer respective antenna datasheet for the antenna keep-out area recommendation.

HOST PCB LAND PATTERN &KEEP-OUT FOR BASE BOARD ANTENNA



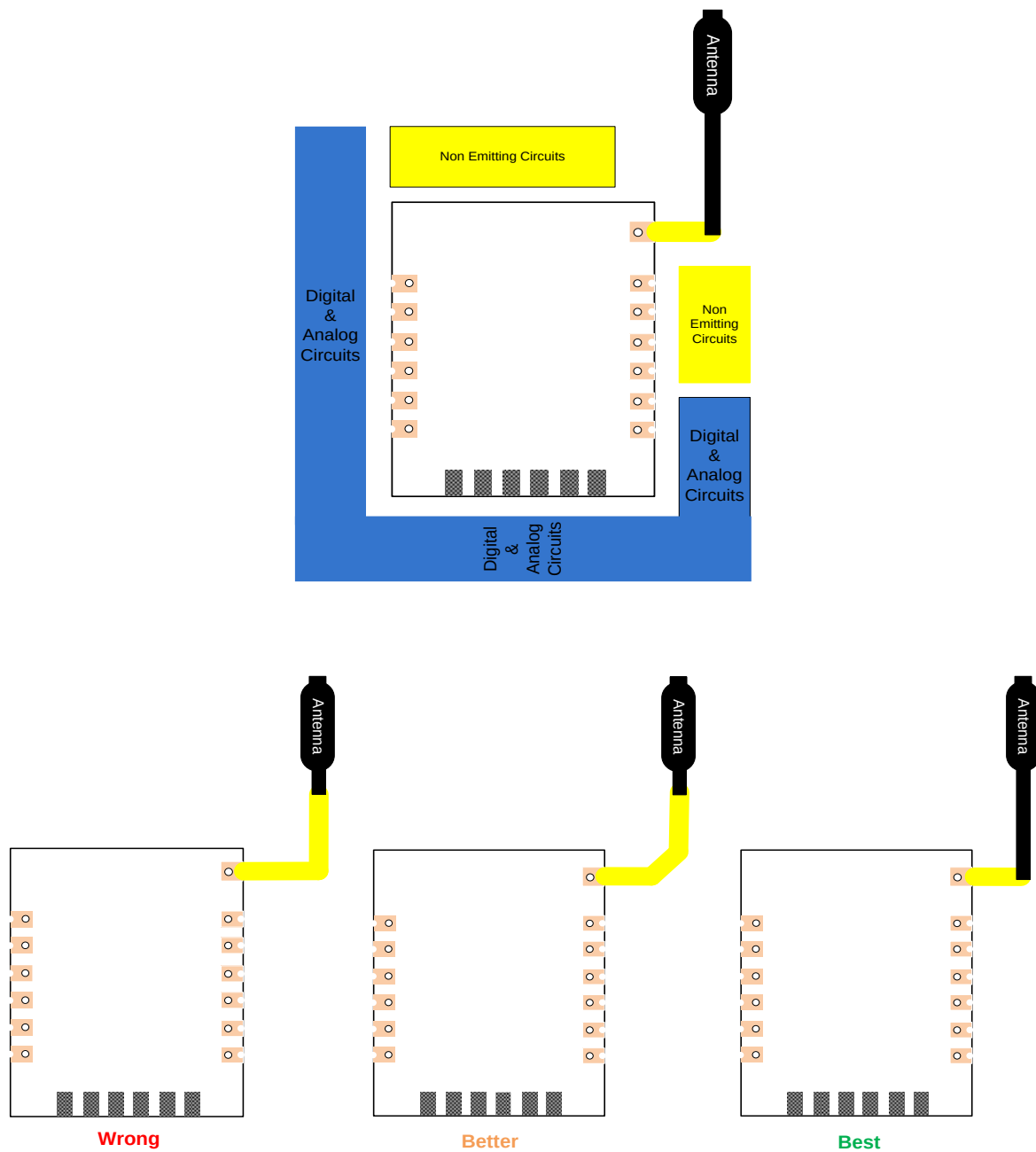
Note:

- Ensure there is no copper in the antenna keep-out area on any layers of the host PCB.
- Put the antenna away from the obstacles.
- Refer respective antenna datasheet for the antenna keep-out area recommendation.

LAYOUT GUIDELINES (EXTERNAL ANTENNA)

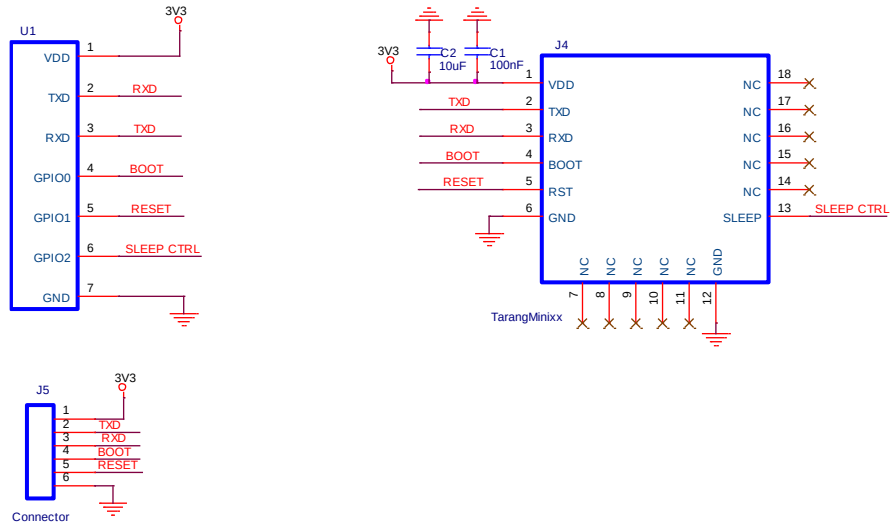
Placement and PCB layout are critical to optimize the performance of a module without on-board antenna. The trace from the antenna port of the module to an external antenna should be 50Ω and it must be as short as possible to avoid any interference into the transceiver of the module. The location of the external antenna RF port of the module should be kept away from any noise sources and digital traces.

As indicated in picture below, RF critical portion of the module should be clearly separated from any digital circuits on the system board.



8. OTHER DESIGN NOTES

TYPICAL INTERFACE WITH A HOST CONTROLLER

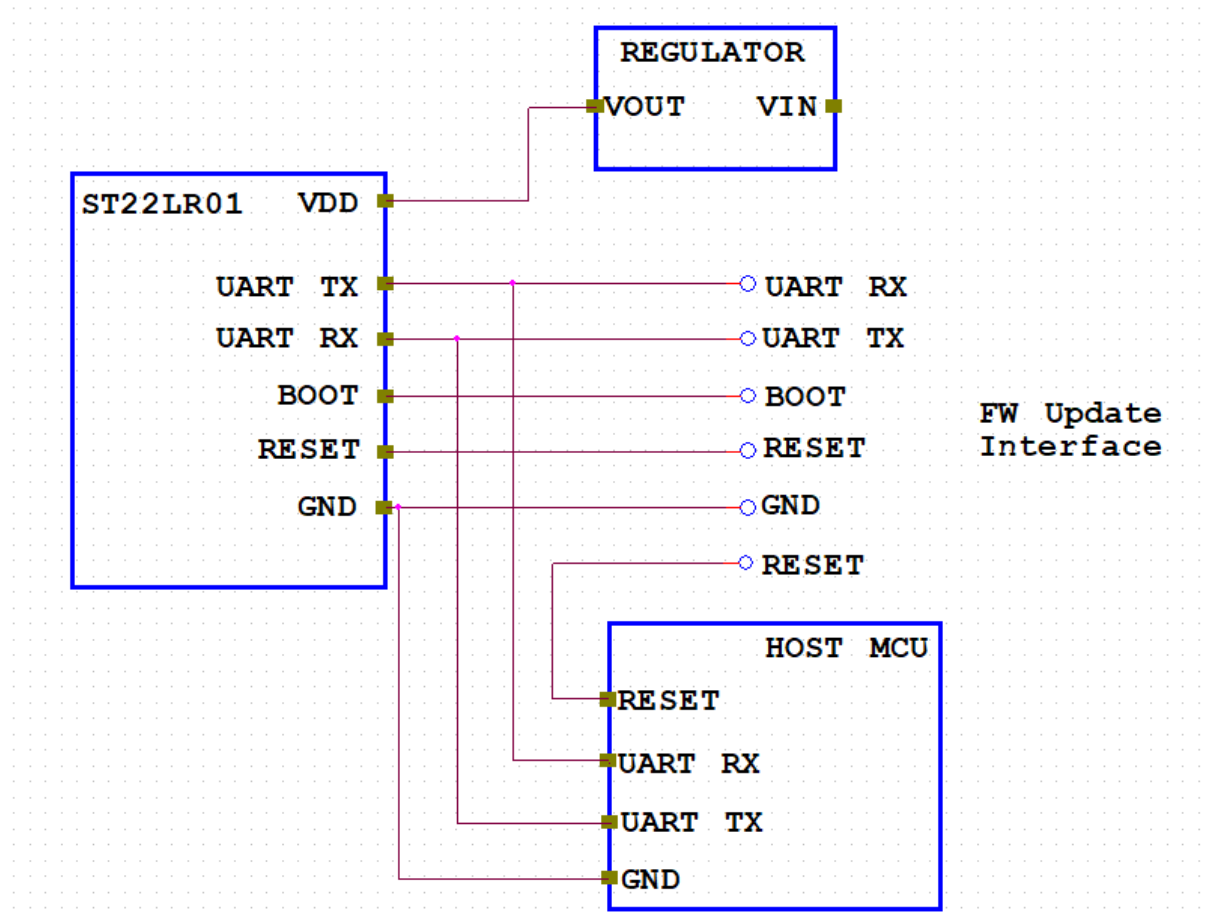


Notes:

1. **U1** represents host MCU. Power domain of GPIO interfaces on the module is 3V3 (Typ). Level translation circuit is not required if the domain on **U1's** GPIO interfaces is the same
2. A level translator should be used if the power domain of the Host controller is different
3. Default firmware upgrade method is through the internal boot-loader of the module via main UART. It is recommended to place a 6-pin (**J5**) connector header on the host PCB with VDD, TXD, RXD, BOOT, RESET & GROUND
4. Put the Host controller into high impedance/tri-state while accessing connector **J5** for external firmware upgrade.
5. All the unused pins should be left floating; do not ground.

FIRMWARE UPDATING RELATED RECOMENDATIONS

To enable firmware updating an external UART interface connection as shown below. ST22LR01 firmware can be updated through the UART interface by holding the host MCU in reset state which typically will free the UART lines to be used by the update interface.



OTHER DESIGN NOTES

A good power supply design is critical for proper device operation. If the supply voltage is not kept within tolerance, or is extensively noisy, it may degrade device performance and reliability. To help reduce noise, we recommend placing both 10uF and 100nF capacitor near to VDD. Make power and ground trace thicker than signal traces and make them able to comfortably support the maximum current specifications. Ground planes are preferable.

- ✓ Avoid test points, exposed pads or vias below the module.
- ✓ Keep this module away from other high frequency circuits
- ✓ This module shouldn't be mechanically stressed when installed.
- ✓ Keep this module away from heat.

IMPROVE ANTENNA PERFORMANCE

Antenna location is important for optimal performance. In general, antenna elements radiate perpendicular to the direction they point.

Metal objects near the antenna cause parasitic coupling and detuning, preventing the antenna from radiating efficiently. Metal objects between the transmitter and receiver can also block the radiation path or reduce the transmission distance, so position Module antenna away from them as much as possible.

9. SOLDERING RECOMMENDATIONS

TarangMini™ ST22LR01 modules are manufactured following standard lead-free reflow profile IPC/JEDEC J-STD-020. This module can be soldered to the host PCB using standard leaded and lead-free solder reflow profiles. Follow the below recommendation to avoid damaging the module.

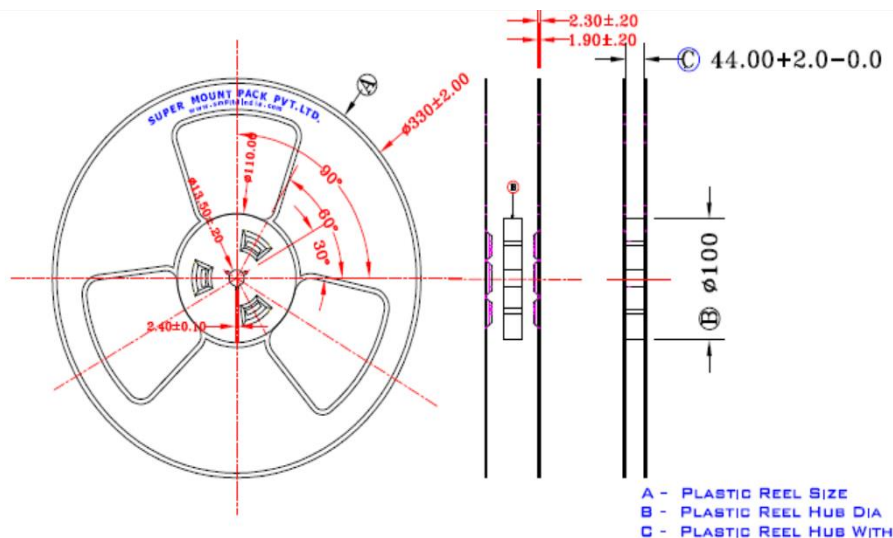
- ✓ Do not exceed peak temperature (235 ~ 245°C)
- ✓ Refer to the solder paste data sheet for specific reflow profile recommendations.
- ✓ Use no-clean flux solder paste.
- ✓ Do not wash as moisture can be trapped under the shield.
- ✓ Use only one flow. If the PCB requires multiple flows, apply the module on the final flow.

10. PACKAGING & SHIPPING INFORMATION

TAPE & REEL PACKAGE INFORMATION

Reel Specifications:

Material: High Impact Polystyrene (HIPS)



DIMENSIONS	Plastic Reel Dia	Plastic Reel Thick	Plastic Hub Dia	Plastic Hub With
SPECS' Nominal	Ø330.00	2.30	Ø100.00	44.00
Tolerance +/-	2.00	0.20	1.00	+2.00/-0.00

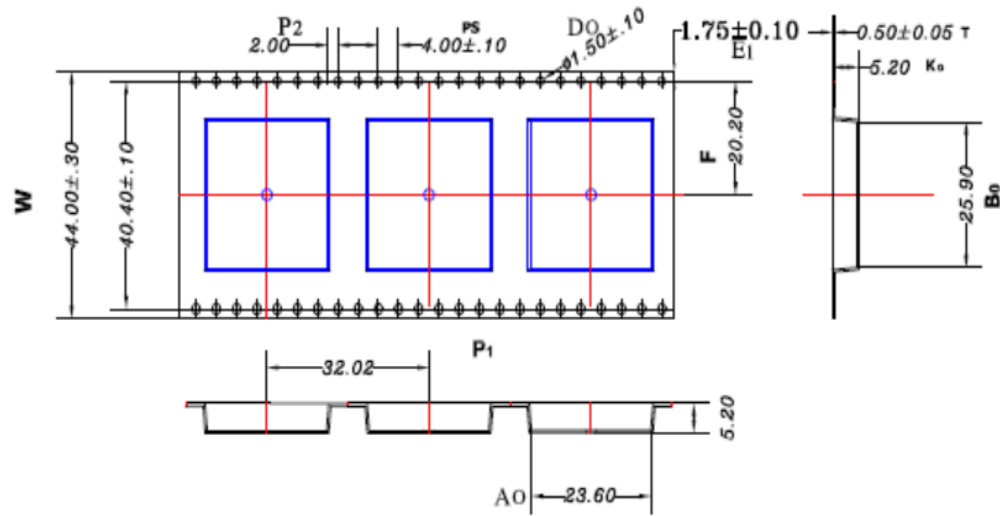
- All dimensions are in mm.

This Certificate - of - Compliance Certifies that the product has been manufactured and inspected in conformance to EIA 481-D Standards.

Tape Specifications:

Material: PS Conductive Black

Surface Resistivity: 10^5 Ohms/Square



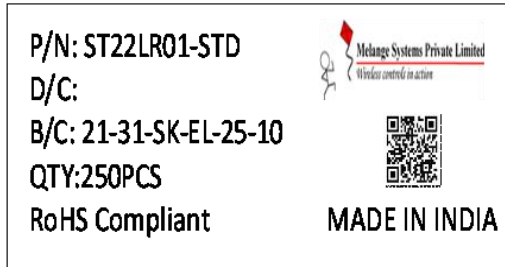
DIMENSIONS	W	Ao	A1	Bo	B1	P0	P1	P2	K0	K1	T	E1	F
SPECS' Nominal	44.00	23.60	N.A	25.90	N.A	4.00	32.02	2.00	5.02	N.A	0.50	1.75	20.20
Tolerance +/-	0.30	0.20	-	0.20	-	0.10	0.10	0.10	0.10	-	0.05	0.10	0.10

- All dimensions are in mm.

This Certificate - of - Compliance Certifies that the product has been manufactured and inspected in conformance to EIA 481-B Standards.

LABELING AND SHIPPING INFORMATION

The following labels are located on the Reel Case.



CARTON CONTENTS



<Pizza Box: 1 Reel per Pizza Box>



< Pizza Box: 1 Reel>



<Master Carton: 8 Pizza Boxes/ctn>

*Please note: for customised module purchase, reel quantity, reel dimensions, and package qualities are subject to change.

11. SUPPORTED FIRMWARE

TARANGCONNECT - LW-S201

Completely integrated LoRaWAN® Binary supporting Class A & Class C operations. Can be controlled by a host controller using simple UART based APIs

TARANGCONNECT - TN-S101

Firmware Binary supporting module to module, point-to-point, point-to-multipoint topologies. Can be controlled by a host controller using simple UART based APIs, also supports transparent mode of operation.

* Please refer respective documentations for more details